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1 Lo7 Summary Report

IIoT Network Analysis: Age of Information and Reliability Trade-offs

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1. Introduction

In this report, I analyze how data freshness and packet delivery reliability compete with each other in mixed-traffic IIoT networks, building on the framework that Farag et al. (2023) developed. The assignment involved exploring a simulated IIoT dataset, building machine learning models to predict AoI, and drawing insights about how network parameters affect data freshness and reliability.

From reading the Farag et al. (2023) paper, I learned that Age of Information (AoI) tracks how many time slots have passed since the controller last received a fresh reading from a sensor. While delay only looks at how long one packet takes to arrive, AoI tells you how outdated the controller's picture of the process actually is. What stood out to me is that AoI matters greatly in IIoT because control systems rely on fresh sensor data to make real-time decisions, such as SCADA controllers monitoring pressure in an oil refinery.

The paper distinguishes two types of traffic that share an unreliable channel, creating a fundamental tradeoff. AoI-oriented traffic consists of periodic monitoring readings where freshness is the goal, following a generate-at-will model. Deadline-oriented traffic consists of sporadic safety-critical alarms that must be delivered within a strict time bound D , with reliability measured by Packet Loss Probability (PLP) (Farag et al., 2023).

2. Data Exploration Findings

The dataset contains 10,000 observations with 1,397 infinite AoI values (about 14%), representing cases where sensor updates never reached the controller. Based on my exploration, three key patterns emerged:

Capture threshold drives PLP: A moderate positive correlation (0.24) exists between γ and PLP, since higher thresholds require stronger SINR for successful decoding.

Network density is the strongest PLP predictor: `num_nodes` shows a 0.56 correlation with PLP, reflecting increased contention on the shared channel as more nodes compete.

Channel quality inversely affects PLP: A -0.40 correlation between `channel_quality` and PLP confirms that better channel conditions improve packet delivery reliability.

What surprised me is that none of the individual parameters showed a strong link to AoI (all below 0.07). This tells me that AoI is not driven by any one factor on its own, but by the random nature of multiple sensors trying to send data at the same time over a shared channel.

3. Machine Learning Results

A Random Forest regressor was trained on four features (transmission probability, capture threshold, number of nodes, channel quality) to predict AoI. The negative R-squared of -0.5548 told me right away that the model struggles with AoI prediction. The heavy-tailed distribution of AoI, with values spanning from 1 to over 52,000 time slots, makes point prediction extremely difficult from these four features alone.

What I find most valuable despite the poor overall fit is that the feature importance rankings are meaningful: capture threshold (44%), number of nodes (32%), channel quality (15%), and transmission probability (9%). This ranking aligns with the SINR-based analytical model in Farag et al. (2023).

For the Bonus Challenge, I built a TensorFlow neural network predicting both AoI and PLP simultaneously. I found it interesting that the neural network achieves a slightly positive R-squared for AoI (0.0181 vs -0.5548 for RF) and a much stronger R-squared of 0.6131 for PLP. This tells me PLP is much more predictable from these four features than AoI is, which makes sense given the stronger correlations I saw in the heatmap.

Metric	Random Forest (AoI)	Neural Network (AoI)	Neural Network (PLP)
MSE	~3,423,000	~2,162,000	~0.012
R ²	-0.55	0.02	0.61

4. Optimization Strategies

Strategy 1 - Adaptive access control: Dynamically adjust transmission probabilities based on the deadline-oriented queue state. When the TD queue is empty, monitoring nodes transmit aggressively to reduce AoI; when safety packets are queued, monitoring nodes back off to protect PLP.

Strategy 2 - Receiver sensitivity improvement: Deploying receivers with lower capture thresholds fundamentally eases the tradeoff. At gamma = -2 dBm, the paper showed AoI remains nearly flat regardless of TD transmission activity.

5. Real-World Applications

Drawing from my background in Oil and Gas, I can see direct applications of these concepts.

Oil and gas pipeline monitoring: Balancing AoI for routine flow/pressure readings against PLP for emergency leak alarms ensures both predictive maintenance and safety compliance.

Smart grid distribution: Fresh voltage readings (low AoI) enable grid optimization, while reliable fault signals (low PLP) ensure automatic protection devices respond within safety deadlines.

6. Conclusion

This assignment gave me a deeper appreciation for the complexity of managing competing traffic types in IIoT networks. The three factors that most influence the AoI-PLP tradeoff are capture threshold, number of contending nodes, and channel quality. My machine learning analysis reinforced this finding, with capture threshold alone accounting for 44% of feature importance in AoI prediction.

One of the most useful takeaways is that PLP is far more predictable from network parameters than AoI is (R-squared of 0.6131 vs 0.0181), because AoI depends heavily on random channel events that static features cannot capture. This suggests that in practice, engineers can more reliably plan for deadline-oriented reliability than for information freshness, and should consider adaptive access control or receiver upgrades to manage AoI effectively.

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